Re: Freight truck TCD paper -- a few questions

Yip, Arthur <Arthur.Yip@nrel.gov>
Mon 2024-07-01 4:13 PM
To:Dwyer, Michael (CONTR) (EIA) <michael.dwyer@eia.gov>

Also, I am surprised the sales by region from macro, matched with the new regional survival curves, don't create more problems in regional stocks (trend over time), but maybe your calibration based on age is working that out.

Do your sales and stocks by region trend in a stable manner, or fluctuate, as the new regional survival curve makes its way through the existing stock (which has its own quirks in age distribution i.e. the recessions, COVID, etc.)?

Arthur

From: Yip, Arthur <Arthur.Yip@nrel.gov>
Sent: Friday, June 28, 2024 1:41 PM
To: Dwyer, Michael (CONTR) (EIA) <michael.dwyer@eia.gov>
Subject: Re: Freight truck TCD paper -- a few questions

Thanks Michael - I think you're right in that vehicles of all powertrains get scrapped (and added when SSURV > 1) at the same rate as gasoline / any type of vehicle in each region - but I think there's a missing mechanism of how the powertrain mix of transferred vehicles can look different from the existing stock.

Since survival is applied to the stock of every powertrain, the application of SSURV (when <1) scraps vehicles according to the existing stock in the region, e.g. drop 3% of previous year's ICEVs, 3% of previous year's BEVs, etc.

And when SSURV is >1, the model also adds vehicles, according to the existing stock in the region, e.g. add 4% of previous year's age X ICEVs, 4% of previous year's age X BEVs.

However, while the total number lines up (4% more total vehicles), there are possibilities that the same amount of stock "addition" could look more like +2% ICEVs, +20% BEVs, depending on ICEV / BEV stock numbers, what the used vehicle market is looking like, etc.

Essentially, when EVs are removed or added in one region, there isn't a link to accounting whether those vehicles exist or kept in balance (and I agree there is no easy way to do this without explicit modeling of movements, or perhaps some sort of parallel/sequential modeling of regions). My concern is that because adoption is so uneven, EVs could be being removed at large quantities in the CA-containing Pacific region (proportional to the stocks by type in each region), and but not being added in large enough quantities in the non-CA-containing regions.

Curious how this looks in the results of LDV stock by technology type and by region.

Thoughts?

I don't know if technology & region -specific survival curves would help much either (assuming one could even get enough good data on that), and rather, that might just solidify current near-term trends into the modeling, rather than appropriately project some designed future scenario of vehicle movements.

Arthur

From: Dwyer, Michael <Michael.Dwyer@eia.gov>
Sent: Friday, June 28, 2024 6:09 AM
To: Yip, Arthur <Arthur.Yip@nrel.gov>
Subject: RE: Freight truck TCD paper -- a few questions

CAUTION: This email originated from outside of NREL. Do not click links or open attachments unless you recognize the sender and know the content is safe.
Hi Arthur,

No problem, I thought it was interesting and hope to have more discussions in the future.

We don't explicitly "remove" vehicles from **region x** and "add" vehicles to **region y**. The curves are applied directly to the previous year's stock for every powertrain (index *ildv*). See the code below (SSURV25 is the *annual* survival – i.e., the ratio of cumulative survival in vintage *y* and that in vintage *y-1*).

```
do iregn = 1,mnumcr-2
  do IVTYP = 1,maxvtyp
    do ILDV=1,maxldv
      do ihav = 1,maxhav
        if(ihav.eq.1) LDV_STOCK(iregn,ivtyp,1,ILDV,1,ihav,n) = sum(nldvsales(iregn,ivtyp,1:maxclass,ildv,n))
        do iage=2,maxage-1
          LDV_STOCK(iregn,IVTYP,1,ILDV,iage,ihav,n) = LDV_STOCK(iregn,IVTYP,1,ILDV,iage-1,ihav,n-1)*SSURV25(iregn,iage-1,
        enddo
        LDV_STOCK(iregn,IVTYP,1,ILDV,maxage,ihav,n) = LDV_STOCK(iregn,IVTYP,1,ILDV,maxage-1,ihav,n-1)*SSURV25(iregn,maxage
                                                      LDV_STOCK(iregn,IVTYP,1,ILDV,maxage,ihav,n-1)*SSURV25(iregn,maxage,
      enddo
      transfer retired fleet stock to non-fleet stock (taxis are not transfered to HH stock)
      do iage=1,maxage
        LDV_STOCK(iregn,IVTYP,1,ILDV,iage,1,n) = LDV_STOCK(iregn,IVTYP,1,ILDV,iage,1,n) + OLDFSTKT(iregn,IVTYP,ILDV,iage)
      enddo
    enddo
  enddo
enddo
```

In other words, we currently assume that vehicles of all powertrains move about the country and are scrapped at the same rate as gasoline vehicles.

--Michael

From: Yip, Arthur <Arthur.Yip@nrel.gov>
Sent: Thursday, June 27, 2024 5:01 PM
To: Dwyer, Michael <Michael.Dwyer@eia.gov>
Subject: Re: Freight truck TCD paper -- a few questions

Hi Michael,

Thanks for the discussion last week.

In the regions and ages where you have >=1 survival, when you add used vehicles of a certain age into the fleet, how do you figure its powertrain mix?
Same mix as the existing stock?
Or do you run these additions through the technology choice module? Perhaps a "used vehicle choice model" ?
And/Or do you have some parallel national system where you can pull from some stockpile of discarded national stock and/or the fleet stock?

Arthur

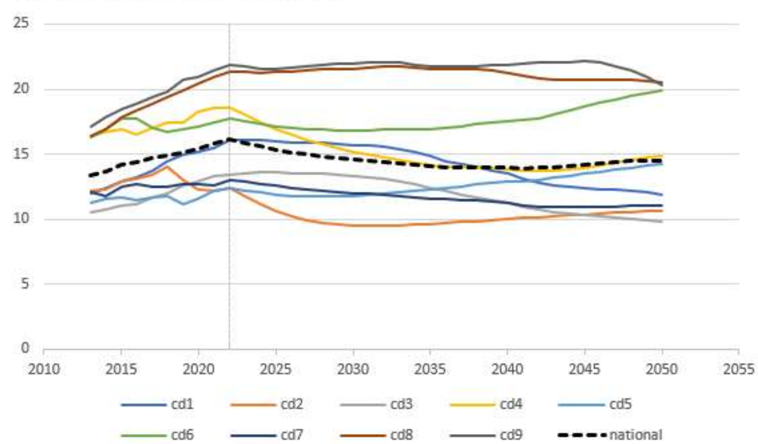
From: Dwyer, Michael <Michael.Dwyer@eia.gov>
Sent: Thursday, June 13, 2024 7:44 AM
To: Yip, Arthur <Arthur.Yip@nrel.gov>; Ledna, Catherine <Catherine.Ledna@nrel.gov>; Jadun, Paige <Paige.Jadun@nrel.gov>
Cc: Muratori, Matteo <Matteo.Muratori@nrel.gov>; Birky, Alicia <Alicia.Birky@nrel.gov>
Subject: RE: Freight truck TCD paper -- a few questions

CAUTION: This email originated from outside of NREL. Do not click links or open attachments unless you recognize the sender and know the content is safe.
Hi Arthur,

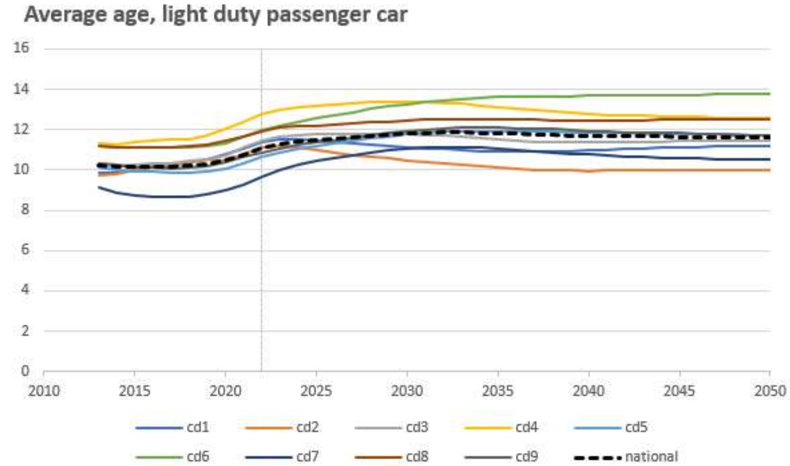
June 20 and 21 are both good, our schedules are pretty flexible those days.

Yes this approach results in different average ages by region, e.g. Class 8 tractors and LDV passenger cars below.

Average vehicle age - C8 tractor



Average age, light duty passenger car



I calibrated the regional curves (9 regions, 11 truck size classes / vehicle types, 2 LDV types) so that the national average age in 2050 for each size class matches (+/- 1%) that produced by NEMS if national scrappage is applied for all regions.

Our national truck sales projection (from macro) is divvied out to regions and truck types/classes based on the regional distribution in our last historical year of data. For LDVs, national sales are from macro and we project the regional sales distribution based on the change in projected licensed drivers.

For light duty, NEMS TRAN has historically transferred fleet vehicles to household as they age. For AEO2023 I updated these transfer rates using the difference between *fleet* scrappage rates and national [household+fleet] scrappage rates. This is particularly important for rental fleets, which dump nearly 100% of their vehicles by the time they are 3 years old.

That methodology makes sense to me. One slight definitional difference: we don't have a "vintage 0" – sales are vintage 1 and it goes on up from there. I like *conditional* as a label since it hints at how survival is basically a probability. I tend to call that *annual survival*, since it's YoY change, as opposed to *cumulative survival* which is the product of all the previous *annual*s.

--Michael

Michael Dwyer
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From: Yip, Arthur <Arthur.Yip@nrel.gov>
Sent: Monday, June 10, 2024 5:58 PM
To: Dwyer, Michael <Michael.Dwyer@eia.gov>; Ledna, Catherine (NREL) <catherine.ledna@nrel.gov>; Jadun, Paige (NREL) <paige.jadun@nrel.gov>
Cc: Muratori, Matteo (NREL) <matteo.muratori@nrel.gov>; Birky, Alicia (NREL) <alicia.birky@nrel.gov>
Subject: Re: Freight truck TCD paper -- a few questions

Hi Michael,

Finally looking to put this on the calendar - how about Fri Jun 14 11 AM? Curious to hear how your regional LDV and MHDV survival curve implementation is going.
In the simulations, do you achieve different average ages by region? Different stock numbers by 2050 in each region? Can you validate resulting average ages with observed ages, for example? Do new vehicle sales still get determined at a national level, or have those also been regionalized?

To answer your questions:
We've combined household + fleet together so far for these survival curve analyses, but now that I'm reviewing old emails, I see that you/EIA were "still using national-average survival curves for vehicles that remain in the fleet stock." But fleet vehicles likely have their own survival curve? And

Class 1-2a vehicles only here, so far.

The following are the equations David Greene and I agreed on, for our analysis of regional survival (see exactly that recent paper on scrappage and vehicle lifetimes. for more https://baker.utk.edu/wp-content/uploads/2023/03/Formatted-Vehicle-Scrappage-and-Survival_Single-Column_7Mar23.pdf)

Define age as: (calendar year - model year).
Let n(t,a,c) be the number of vehicles in operation of vehicle class & type c, of age a, in calendar year t.
The conditional survival probability is p(a | a-1) = n(t,a,c)/n(t-1,a-1,c).
To get the vehicle survival curves that we use in modeling (p_t(a) unconditional cumulative probability density function that represent the probability of surviving to a given age, a, for a new vehicle sold in year t, this can be computed from the cumulative product of the conditional survival probabilities.
p_t(a) = p(a|a-1) * p(a-1|a-2) * p(a-2|a-3) * ... * p(1|0)

From: Dwyer, Michael <Michael.Dwyer@eia.gov>
Sent: Wednesday, May 1, 2024 6:56 PM
To: Yip, Arthur <Arthur.Yip@nrel.gov>; Ledna, Catherine <Catherine.Ledna@nrel.gov>; Jadun, Paige <Paige.Jadun@nrel.gov>
Cc: Muratori, Matteo <Matteo.Muratori@nrel.gov>; Birky, Alicia <Alicia.Birky@nrel.gov>
Subject: RE: Freight truck TCD paper -- a few questions

Hi Arthur,

To clarify – the plotted survival curves are *cumulative* not *annual*. They were derived from the average *annual* survival across multiple years.

The regional curves don't produce the *exact* same stocks as the national survival when applied. I'm still working on calculating and documenting the exact delta for that. But I wouldn't expect them to, given the fact that I fit curves to each (and smoothed out historical anomalies, e.g. GFC). We want a reasonable guess at future survival rates rather than an exact fit to historical behavior.

Nice work on the state survival rates – is that household only or household + fleet? And does it include all pickups or only Class 1-2a?

Speaking of David Greene, I was considering an attempt to include some of the trend in vehicle longevity discussed in his recent paper on scrappage. Just need 10 more FTEs on the transportation team 😊

--Michael

From: Yip, Arthur <Arthur.Yip@nrel.gov>
Sent: Wednesday, May 1, 2024 6:36 PM
To: Dwyer, Michael <Michael.Dwyer@eia.gov>; Ledna, Catherine (NREL) <catherine.ledna@nrel.gov>; Jadun, Paige (NREL) <paige.jadun@nrel.gov>
Cc: Muratori, Matteo (NREL) <matteo.muratori@nrel.gov>; Birky, Alicia (NREL) <alicia.birky@nrel.gov>
Subject: Re: Freight truck TCD paper -- a few questions

Hey Michael,

Thanks for the preview! It does sound like you have 2 modeling systems to develop/maintan/manage now, the old and the new...

Yes, regional survival, LDV and MHDV, is of significant interest to us, especially as it approximates used vehicle movements. Alicia is also interested, of course. Maybe not this week, but I'm sure some of us would be interested in discussing further in the next few weeks. I'll look into putting something on the calendar.

Does the national and regional modeling line up with your methodology e.g. projected sum of regional stocks = projected national stocks?

We are also about to work with David Greene to dig deeper into regional survival curves for light-duty. Sneak peek / preliminary results attached. Motivated by your work from last year!

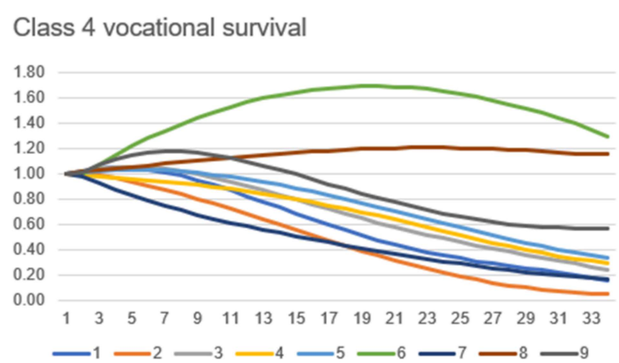
Arthur

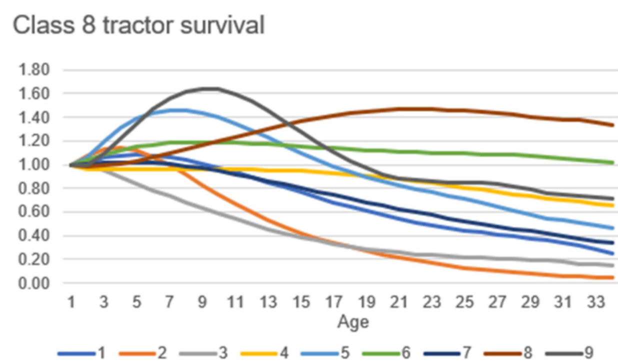
From: Dwyer, Michael <Michael.Dwyer@eia.gov>
Sent: Monday, April 29, 2024 7:46 AM
To: Ledna, Catherine <Catherine.Ledna@nrel.gov>; Jadun, Paige <Paige.Jadun@nrel.gov>; Yip, Arthur <Arthur.Yip@nrel.gov>
Cc: Muratori, Matteo <Matteo.Muratori@nrel.gov>
Subject: RE: Freight truck TCD paper -- a few questions

Catherine/Arthur (and Matteo/Paige if interested),

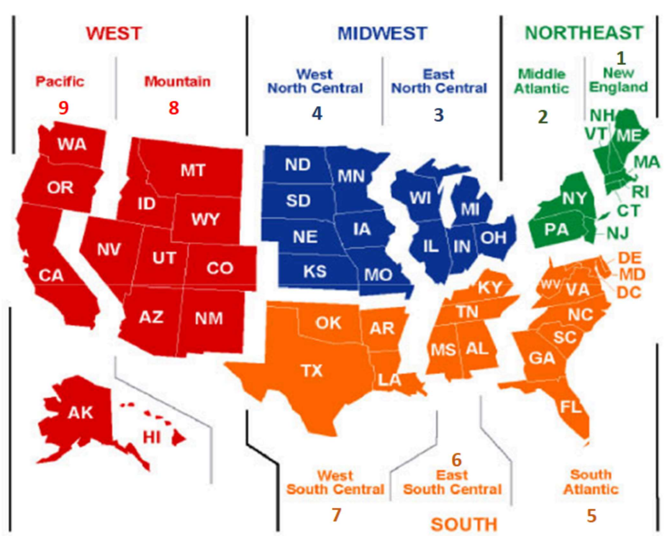
Apologies for the slow follow-up – "skipping" an AEO year resulted in us being *more* busy, not *less* 😊

I used our vehicle registration data (purchased from S&P/Polk), which we have annually from CY2018-CY2022 (will have CY2023 in Aug/Sep). I developed conditional – or "annual" (as opposed to cumulative) -- survival rates based on the average YoY survival of each vintage, by region. What pops out isn't just a survival rate – it also implicitly includes regional transfers since the survival rates can go above 100%. So it captures both survival/scrappage as well as vehicle flows around the country. There are lots of interesting nuggets – for example, Region 6 absorbs a huge amount of used trucks (Class 2b-7, see Class 4 in left chart below). And while Region 9 buys some new Class 8 tractors, it gets most of its vehicles used from other regions (see chart on right).





Regions (sorry, line colors above don't match this)



I used an identical methodology for the light duty vehicles. This is important for NEMS because it allows us to capture how stock fuel economy varies by region, rather than applying a national average to the whole U.S. fleet. Happy to have a discussion sometime this week if you all are available. It would be relatively informal – the model development cycle here is too fast for me to be able to develop presentations, white papers, or methodology deep-dives on any of the work unfortunately.

--Michael

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Lead Transportation Modeler
Office of Long Term Energy Modeling
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From: Ledna, Catherine <Catherine.Ledna@nrel.gov>
Sent: Wednesday, April 17, 2024 4:37 PM
To: Dwyer, Michael <Michael.Dwyer@eia.gov>; Jadun, Paige (NREL) <paige.jadun@nrel.gov>; Yip, Arthur (NREL) <arthur.yip@nrel.gov>
Cc: Muratori, Matteo (NREL) <matteo.muratori@nrel.gov>
Subject: RE: Freight truck TCD paper -- a few questions

Hi Michael,

Yes, that is a correct interpretation of that scenario. We also have additional scenarios (Cons Tech and Adv ICEV Tech) that look at different rates of technology progress for ZEVs and ICEVs.

I'd be interested in hearing more about your work on regional survival curves for freight trucks and the data sources you used. Did you leverage the new VIUS? Happy to join your call with Arthur if useful.

Thanks,
Catherine

From: Dwyer, Michael <Michael.Dwyer@eia.gov>
Sent: Wednesday, April 17, 2024 2:32 PM
To: Jadun, Paige <Paige.Jadun@nrel.gov>; Ledna, Catherine <Catherine.Ledna@nrel.gov>; Yip, Arthur <Arthur.Yip@nrel.gov>
Cc: Muratori, Matteo <Matteo.Muratori@nrel.gov>
Subject: RE: Freight truck TCD paper -- a few questions

Hi all,

@Yip, Arthur (NREL) – sure, feel free to email specific questions, or let me know a few times that might work for you and we can have a call to discuss. I recently developed regional survival curves for freight trucks as well (11 different curves for different size class groups, for each region). They will be used for AEO2025.

@Ledna, Catherine (NREL) – thanks for clarifying. So that implies that the *Central* case is assuming DOE targets are successfully met for BEVs/FCVs but not ICEV/HEVs – is the thinking that the *Central* case represents a future where manufacturers are "all in" on ZEVs?

@Jadun, Paige – Anything you have on how those cost assumptions were developed would be helpful, if you are able to share. E.g. infrastructure cost + amortization, expected demand charges and $/kWh rates, etc. We're working on updating transportation electricity costs in NEMS.

Thanks for the quick responses, and for the interesting paper!

--Michael

From: Jadun, Paige <Paige.Jadun@nrel.gov>
Sent: Wednesday, April 17, 2024 2:52 PM
To: Ledna, Catherine (NREL) <catherine.ledna@nrel.gov>; Yip, Arthur (NREL) <arthur.yip@nrel.gov>; Dwyer, Michael <Michael.Dwyer@eia.gov>
Cc: Muratori, Matteo (NREL) <matteo.muratori@nrel.gov>
Subject: RE: Freight truck TCD paper -- a few questions

Hi Micheal,

We're hoping to publish the charging cost work sometime in the next few months. If helpful, we can send some more info on key assumptions and methods for developing our assumptions.

Thanks!
Paige

From: Ledna, Catherine <Catherine.Ledna@nrel.gov>
Sent: Wednesday, April 17, 2024 1:45 PM
To: Yip, Arthur <Arthur.Yip@nrel.gov>; Dwyer, Michael (CONTR) (EIA) <michael.dwyer@eia.gov>; Jadun, Paige <Paige.Jadun@nrel.gov>
Cc: Muratori, Matteo <Matteo.Muratori@nrel.gov>
Subject: RE: Freight truck TCD paper -- a few questions

Hi all,

Yes, I can confirm that Central = Low Autonomie techs for ICEV/HEVs. Happy to answer any other questions!

Best,
Catherine

From: Yip, Arthur <Arthur.Yip@nrel.gov>
Sent: Wednesday, April 17, 2024 11:39 AM
To: Dwyer, Michael (CONTR) (EIA) <michael.dwyer@eia.gov>; Jadun, Paige <Paige.Jadun@nrel.gov>; Ledna, Catherine <Catherine.Ledna@nrel.gov>
Cc: Muratori, Matteo <Matteo.Muratori@nrel.gov>
Subject: Re: Freight truck TCD paper -- a few questions

Hey Michael,

Always happy to discuss, and separately, I'd love to chat a bit more about what you guys are doing with LDV regional survival curves and MHDV survival. About your questions about our recent published paper,

1) @Jadun, Paige is leading the *MHDV Corridor and Depot Charging Costs* work and provide guidance on how those were produced and when it will be published.
2) I believe so (Central = Low Autonomie tech for ICEV/HEV), but @Ledna, Catherine can confirm that.

Arthur

From: Dwyer, Michael <Michael.Dwyer@eia.gov>
Sent: Tuesday, April 16, 2024 8:04 PM
To: Yip, Arthur <Arthur.Yip@nrel.gov>
Subject: Freight truck TCD paper -- a few questions

Hi Arthur,

I hope you are doing well. I had a chance to go through the MHDV TCD paper you co-authored w/ your NREL/DOE colleagues. I had a couple of quick questions – no rush at all, just trying to make sure I understand the assumptions:

1. Where are the assumed *Central* case charging costs from? There is mention of a paper -- *NREL (Forthcoming). MHDV Corridor and Depot Charging Costs* – which sounds like it will have some more detail on how those costs were estimated. Do you know when that will be published?
2. I didn't see anything on the assumed ICEV/HEV efficiency improvement in the *Central* case. The *Adv. ICEV & HEV Tech Progress* case assumes Autonomie's *High* case, which implies that the *Central* case uses the *Low* tech pathway for ICEV/HEV? Is that correct?

Let me know if I should message the listed lead contact instead, happy to send another email.

Thanks
Michael

eia
Michael Dwyer
Lead Transportation Modeler
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